

Familiarisation with TPS

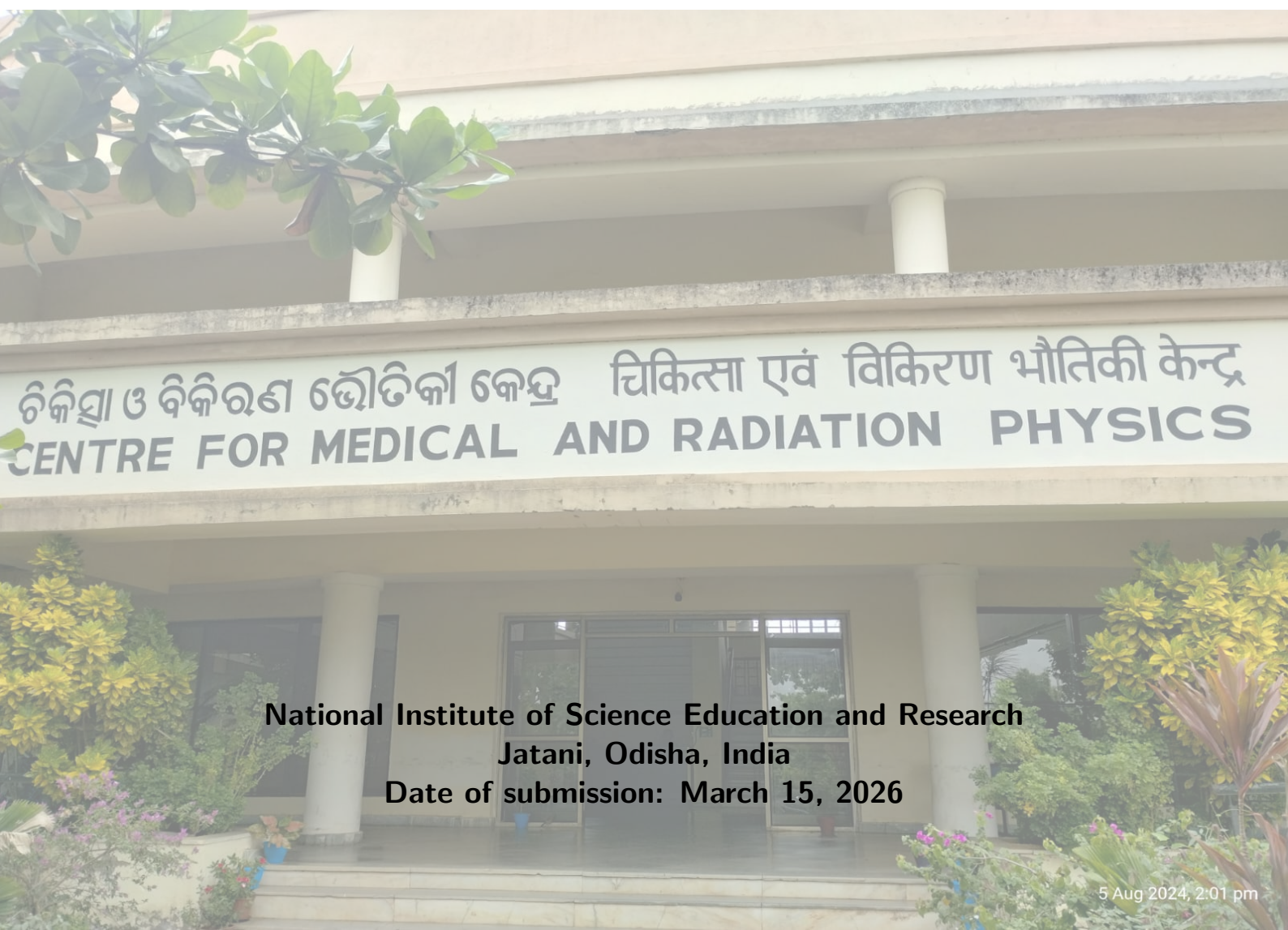


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1 Objective

1. Create a radiotherapy treatment plan using a computerised TPS (Treatment Planning System).
2. Perform treatment planning using various modes (e.g., photons, protons, etc.) of treatment and using different setups of beams to find the best treatment plan.

2 Apparatus

- A computer with MatRad software.
- DICOM image of the Patient.

matRad is an open-source treatment planning system (TPS) research and education toolkit developed in MATLAB for external beam radiotherapy planning [1]. It supports the complete basic planning workflow, including DICOM data import, structure handling, dose calculation, inverse planning, and plan evaluation in a graphical user interface. A major strength of matRad is that it allows planning for different modalities such as photons, scanned protons, and scanned carbon ions within a common framework, which makes it useful for comparative learning.

In this experiment, matRad was used as the primary software platform to understand TPS operation, starting from patient image loading to beam setup and optimization. The software enables users to visualize target volume coverage and dose to organs at risk (OARs), and to assess plan quality through dose distributions and dose-volume histograms (DVHs). Because the code is open and modular, it is particularly suitable for academic use, where both clinical concepts and underlying computational methods can be studied together [1].

3 Theory

4 Observation

5 Conclusion

References

- [1] H.-P. Wieser, E. Cisternas, N. Wahl, S. Ulrich, A. Stadler, H. Mescher, L.-R. Müller, T. Klinge, H. Gabrys, L. Burigo, A. Mairani, S. Ecker, B. Ackermann, M. Ellerbrock, K. Parodi, O. Jäkel, and M. Bangert, “Development of the open-source dose calculation and optimization toolkit matRad,” *Medical Physics*, vol. 44, no. 6, pp. 2556–2568, 2017.